SUSTAINABLE CONCRETE USING FOUNDRY SAND

Experimental Study on Partial Replacement of Fine Aggregate

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Project Description:
Experimental investigation on M25 grade concrete using foundry sand as a partial replacement for natural fine aggregate (0%, 20%, 40%) to evaluate strength, workability, and sustainability.

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Jodhpur, Rajasthan, India

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ABSTRACT

This study investigates the potential of **foundry sand as a sustainable partial replacement for natural fine aggregate** in M25 grade concrete. The increasing depletion of natural river sand and the environmental challenges associated with the disposal of industrial waste have necessitated the exploration of alternative materials in concrete production. Foundry sand, a by-product of the metal casting industry, was utilized as a replacement material at proportions of 0%, 20%, and 40% by weight of fine aggregate.

Experimental analysis was conducted to evaluate the performance of concrete in terms of **workability and compressive strength**. The slump test was performed to assess the workability of fresh concrete, while compressive strength tests were carried out at curing periods of 7, 14, and 28 days. The results indicated that workability decreased with an increase in foundry sand content due to its finer particle size and higher water absorption.

The compressive strength results revealed that the **20% replacement mix (FS20) achieved the highest strength of 31.41 MPa at 28 days**, outperforming the conventional mix while maintaining acceptable workability. However, higher replacement levels showed reduced performance due to decreased cohesion and workability.

The study concludes that **20% replacement of fine aggregate with foundry sand is optimal**, providing a balance between strength, workability, and sustainability. This research demonstrates that foundry sand can be effectively utilized as an **economical, eco-friendly alternative**, contributing to waste management and conservation of natural resources in the construction industry.

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LIST OF ABBREVIATIONS

S. No.	Abbreviation	Full Form
1	FS	Foundry Sand
2	OPC	Ordinary Portland Cement
3	MPa	Megapascal
4	KN	Kilonewton
5	CTM	Compression Testing Machine
6	IS	Indian Standard
7	BIS	Bureau of Indian Standards
8	M25	Concrete Mix with Characteristic Strength of 25 MPa
9	ASTM	American Society for Testing and Materials
10	Kg	Kilogram
11	mm	Millimeter
12	%	Percentage
13	w/c	Water-Cement Ratio
14	SCM	Supplementary Cementitious Material
15	N/mm^2	Newton per Square Millimeter
16	EIA	Environmental Impact Assessment
17	WRA	Water Reducing Admixture
18	F.A.	Fine Aggregate
19	C.A.	Coarse Aggregate
20	OPC 43	Ordinary Portland Cement of Grade 43
21	PPC	Portland Pozzolana Cement
22	RMC	Ready-Mix Concrete

CHAPTER-1
INTRODUCTION

1.1 Background:

Concrete is the most widely utilized building material globally, the backbone of contemporary infrastructure. Its global popularity is due to its high compressive strength, durability, shape ability into any form, and affordability. The major components of concrete are cement, coarse aggregates, fine aggregates (usually river sand), and water. Of these, fine aggregate is important because it occupies voids between coarse particles, adds to strength, and provides adequate workability of the mix.

With accelerated urbanization and rising demand for infrastructure, the use of natural resources, particularly river sand, has dramatically increased. Indiscriminate and uncontrolled mining of river sand has resulted in serious environmental degradation, including the reduction of riverbeds, destruction of aquatic life habitats, and depletion of groundwater resources. Moreover, in most areas, natural sand is finding its way out and becoming costly. This emerging issue requires exploring an alternate material to replace natural sand partially or entirely in concrete.

At the same time, numerous sectors produce enormous amounts of waste material that frequently reaches landfills with resulting environmental implications. One waste material is utilized foundry sand, a foundry industry by-product. Foundry sand refers to high-grade silica sand that is utilized as molding sand within foundries. Once used on multiple occasions, its quality lowers, making it no longer safe for additional cast operations. Handling this used sand creates enormous environmental issues for foundry operations.

Using foundry sand in concrete has a twofold advantage—it minimizes the use of natural sand and provides a useful application for industrial waste. Foundry sand is characterized by those factors that might make it as a partial fine aggregate

replacement for use in concrete, including its fine particle size, silica content, and uniformity. But because of the occurrence of residual binders such as clay and chemicals, its behavior in concrete needs to be intensively examined to provide structural integrity and durability.

Over the last few years, engineers and scientists have begun testing the viability of using foundry sand in building construction. There have been several experimental investigations confirming that foundry sand, used in small amounts, can increase some of the properties of concrete without compromising the strength or workability of concrete.

In the sustainability context, the construction sector is slowly moving towards green practices. The use of green building materials and technologies is no longer optional but mandatory.

1.2 Problem Statement:

The over-extraction of natural river sand has caused serious environmental issues, such as ecological imbalance, habitat loss, erosion, and reduction in groundwater levels. In addition, ongoing extraction of river sand is becoming uneconomical in most areas, with increasing costs and restrictions imposed by the government. In contrast, the foundry sector produces enormous amounts of waste sand, a significant portion of which goes untapped and gets dumped in open areas or landfills, contributing to land pollution.

Therefore, there is a pressing need to determine and validate substitute materials that can replace natural sand in concrete to some extent without affecting its performance. Foundry sand, as an abundant industrial waste material with physical properties similar to natural sand, is a potential substitute. However, prior to its adoption on a large scale, extensive studies are required to analyze its impact on the fresh and hardened properties of concrete.

This study focuses on exploring the possibility of using foundry sand to partially replace fine aggregate in M25 grade concrete. Through this study, we hope to

contribute to the efforts of establishing sustainable and eco-friendly construction methods.

1.3 Objective of the Study:

The main aim of this research is to investigate the application of foundry sand as partial replacement for fine aggregate in concrete and evaluate its effect on the performance of M25 grade concrete. The specific research objectives of the research are listed below:

- To find out the workability of fresh concrete mixes with different contents of foundry sand by the slump cone test.
- To test compressive strength of concrete at various curing periods (7, 14, and 28 days) when fine aggregate is partially replaced with foundry sand.
- To compare the performance of the concrete mixes with foundry sand with that of the control mix (0% replacement).
- To find the optimum percentage of replacement of fine aggregate with foundry sand that gives acceptable strength and workability.

1.4 Scope of the Study:

This study involves the partial substitution of used foundry sand for natural fine aggregate in concrete and assessment of its impact on certain properties of M25 grade concrete. Replacement levels considered are 0% (control mix), 20%, and 40% by weight of fine aggregate.

Study Scope is described below:

- Concrete Grade: M25
- Replacement Levels: 0%, 20%, and 40% foundry sand (by weight of fine aggregate)
- Tests Performed: Workability and compressive strength tests at 7, 14, and 28 days of curing.
- Method of Curing: Water curing under normal conditions.

The research is confined to laboratory-scale testing and does not extend to field applications or structural behavior under load conditions. Moreover, the chemical characterization of foundry sand, its leaching potential, and the durability properties of concrete (e.g., sulfate resistance, chloride penetration, or carbonation) are not addressed in this research.

1.5 Significance of the Study:

This study is highly relevant in terms of environmental sustainability, waste minimization, and new material applications in building construction. The principal contributions and values of this work are discussed as follows:

- Environmental Advantages: By minimizing the use of river sand, the research assists in preserving natural ecosystems, avoiding riverbed degradation, and encouraging sustainable mining.
- Conservation of Resources: The use of industrial by-products in concrete can minimize the reliance on non-renewable resources and prolong their availability for future generations.
- Waste Minimization: Used foundry sand, otherwise sent as waste, can be successfully utilized in concrete, thereby conforming to the waste-to-resource approach and promoting circular economy.
- Cost Efficiency: Used foundry sand is usually cheap or free in comparison with river sand and therefore becomes economical for the production of concrete, particularly in areas where river sand is scarce or has a high transportation cost.
- Practical Applicability: In case the study indicates positive results, the results can be applied to develop guidelines for the practical application of foundry sand in construction, especially for general and non-structural concrete works.

CHAPTER-2
STUDY DOMAIN AND DATA

2.1 Overview:

Concrete is amongst the most utilized and widely accepted construction material on the world map. Concrete consists of water, cement, fine aggregate or sand, and coarse aggregate in the form of crushed stone or gravel.

Within recent years, as natural resource depletion has increasingly risen, there have been added motives for investigators and engineers to pursue sustainable and environment-friendly materials instead of typical building materials. This includes industrial waste products like spent foundry sand (UFS) substituting partially in natural aggregates as ingredients in making concrete.

In this research, experimental work is conducted on M25 grade concrete to assess the performance of concrete when natural fine aggregate is replaced partially with used foundry sand.

The replacements were done at 0% (control mix), 20%, and 40% by weight of fine aggregate. Material properties used in the research have been assumed from IS standards and established literature since no laboratory material testing was conducted.

This chapter discusses in detail the materials utilized in the study, their estimated properties according to standards, sources, and the used concrete mix design approach.

2.2 Cement:

Regular Portland Cement (OPC) from 43 Grade as per IS 8112:2013 was utilized in this research. This type of cement is most commonly used and ideal for general building purposes, even including reinforced concrete. The properties are assumed from regular specifications and past studies.

TABLE – 2.1		
Assumed Properties of Cement		
Property	**Assumed Value**	**Source**
Specific Gravity	3.15	IS 4031 (Part 11)
Standard Consistency	29%	IS 4031 (Part 4)
Initial Setting Time	30 minutes	IS 4031 (Part 5)
Final Setting Time	600 minutes	IS 4031 (Part 5)
Fineness	5% retained on 90µm	IS 4031 (Part 1)
28-day Compressive Strength	45 MPa	IS 4031 (Part 6)

2.3 Fine Aggregate (Natural River Sand):

The fine aggregate (natural sand) is taken to be river sand according to Zone II grading as specified by IS 383:2016. River sand is widely applied in construction work because it has a clean texture and sufficient strength. The values are taken from standard sources.

TABLE - 2.2		
Assumed Properties of Fine Aggregate		
Property	**Assumed Value**	**Source**
Specific Gravity	2.64	IS 2386 (Part 3)
Fineness Modulus	2.70	IS 2386 (Part 1)
Grading Zone	Zone II	IS 383:2016
Bulk Density (Loose)	1450 kg/m³	IS 2386 (Part 3)

Fine aggregates play a key role in concrete's workability. The selected sand falls within permissible grading limits.

2.4 Coarse Aggregate:

Crushed granite stone of **20 mm nominal size** is used as coarse aggregate in this study. The aggregates are angular in shape and known to offer better interlocking in concrete.

TABLE – 2.3		
Assumed Properties of Coarse Aggregate		
Property	**Assumed Value**	**Source**
Specific Gravity	2.70	IS 2386 (Part 3)
Water Absorption	0.5%	IS 2386 (Part 3)
Bulk Density (Loose)	1400 kg/m³	IS 2386 (Part 3)
Impact Value	20%	IS 2386 (Part 4)
Crushing Value	25%	IS 2386 (Part 4)

These properties ensure that the aggregates are strong and suitable for structural-grade concrete.

2.5 Water:

The mixing and curing water is supposed to be drinkable tap water with no bad contaminants. The IS 456:2000 specifies that potable drinking water is also qualified to use as a material in making concrete.

2.6 Foundry Sand:

Used foundry sand was used from Mechanical Engineering Foundry Lab at Jodhpur Institute Of Engineering And Technology.

The sand is typically dark gray or black in color, as a result of being mixed with burnt carbon and other additives. It is finer than river sand found naturally and consists of residual binders like carbon and bentonite. Though not chemically

analyzed during this investigation, the foundry sand was noted to be dry, consistent, and can be reused.

TABLE – 2.4		
Assumed Properties of Foundry Sand		
Property	**Assumed Value**	**Source**
Specific Gravity	2.50	Literature
Fineness Modulus	2.10	Literature
Texture	Fine, smooth	Visual Inspection
Shape	Sub-angular	Literature

Foundry sand is rich in silica and has a particle size close to that of natural sand, making it a viable alternative for partial replacement.

2.7 Concrete Mix Design:

The concrete mix is designed as per **IS 10262:2019** for **M25 grade concrete**. The target mean strength is calculated using a standard deviation of 4 MPa. Design Parameters:

- Grade of Concrete: M25
- Target Strength: 31.6 MPa
- Maximum Aggregate Size: 20 mm
- Workability: 75–100 mm
- W/C Ratio: 0.45
- Exposure: Mild
- Cement: OPC 43 grade

TABLE – 2.5	
Mix Proportions (By Weight)	
Component	**Quantity (kg/m³)**
Cement	400
Fine Aggregate	400
Coarse Aggregate	800
Water	180

This mix design is used for the control mix (0% foundry sand). The 20% and 40% replacement mixes were prepared by replacing an equivalent weight of fine aggregate with foundry sand.

TABLE – 2.6						
Mix Variations						
Mix ID	**Foundry Sand (%)**	**Natural Sand (%)**	**Cement (kg)**	**FA (kg)**	**CA (kg)**	**Water (kg)**
FS0	0	100	400	400 (400+0)	800	180
FS20	20	80	400	400 (320+80 FS)	800	180
FS40	40	60	400	400 (240+160 FS)	800	180

Note: FS = Foundry Sand

2.8 Batching, Mixing, and Casting:

All materials were also measured on an electric balance. Cement, foundry sand, and aggregates were dry mixed for 2–3 minutes. The water was slowly added to them, and the whole mixture was mixed well till uniform in consistency.

Standard steel molds with size 150 mm × 150 mm × 150 mm were used in which concrete cubes of the specified sizes were poured. Three layers of each one filled and hammered with a hand-held tamping rod.

2.9 Curing:

After demolding the specimens after 24 hours, they were immersed in a water tank for curing. The cubes were tested for **compressive strength** at 7, 14, and 28 days.

TABLE – 2.7			
Experimental Program Summary			
Mix	**Foundry Sand (%)**	**No. of Cubes**	**Tests Conducted**
FS0	0	9	Slump Test, Compressive Test
FS20	20	9	Slump Test, Compressive Test
FS40	40	9	Slump Test, Compressive Test

Total: **27 cubes + 3 slump tests**

2.10 Assumptions and Limitations:

- All properties of material are taken from IS codes and literature, not determined.
- Chemical analysis and heavy metal content of foundry sand were not investigated.
- Tests of durability (e.g., permeability, shrinkage) were not conducted.
- The project confines to compressive strength and slump test results.

CHAPTER-3
METHODOLOGY

1.1 Overview:

Concrete is amongst the most used construction materials across the world because of its superior structure and durability.

Yet, excessive use of natural resources, particularly river sand, poses serious environmental issues. This research seeks to examine an eco-friendly method through the substitution of natural fine aggregates with used foundry sand (UFS), a metal casting industry waste byproduct.

The technique described here introduces the step-by-step procedures followed in material selection, concrete mix design, sample preparation, curing, and testing methods, adhering to Indian Standard guidelines.

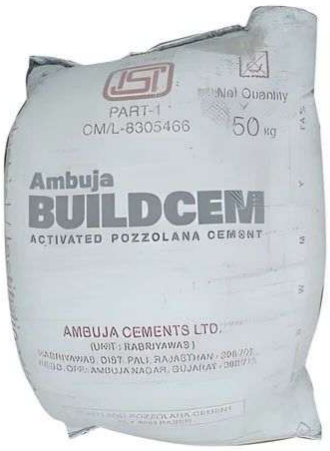
1.2 Research Objective:

The key goals propelling the experimental work are:

- To investigate the workability and compressive strength of M25 grade concrete with the use of foundry sand as partial replacement of fine aggregate.
- To identify the optimal percentage of foundry sand that provides equivalent or improved performance compared to the control mix.
- To assess whether waste foundry sand can be used efficiently in construction, mitigating environmental load.

1.3 Materials Used:

Cement: 43-grade Ordinary Portland Cement (OPC) was utilized during the research. It is in accordance with IS 8112:2013. This cement is suitable for general construction purposes and possesses good early strength for 7,14 and 28-day testing. The used cement was fresh, free from lumps, and kept in a dry environment to prevent moisture absorption.



Fine Aggregate (Natural Sand): River sand as per IS 383:2016 was employed as the fine aggregate. It is of Zone II grading according to the IS classification. The sand was silt-free, clay-free, organic matter-free, or any other impurity-free and was air-dried prior to use.

Coarse Aggregate:

Crushed granite aggregate with a nominal size of 20 mm was used as coarse aggregate, as per IS 383:2016 specifications. They were having a rough texture and angular shape, which enhances bonding with cement paste.

Foundry Sand:

Used foundry sand was sourced from a nearby casting company in Jodhpur. It was dark grey in color, fine textured particles, and sub-angular grain. No chemical tests were conducted, but physical properties were presumed from similar literature surveys. Foundry sand is normally high in silica content and thus can be used as a substitute for sand.

Water:

Clean drinking water from the tap was utilized in the concrete mixes and curing. It contained no organic and inorganic impurities, oils, and salts.

1.4 Mix Design:

Concrete mix design was conducted according to the IS 10262:2019 code standard, which has a scientific methodology of proportioning the concrete components to attain a desired strength and durability for a given grade. In this research, the target concrete grade was M25, which means the characteristic compressive strength of 25 MPa at 28 days.

To maintain safety and adjust for site variability, a target mean strength of around 31.6 MPa was chosen employing the following formula:

$$f_{ck} = f_{ck} + 1.65 \times S$$

Where:

- f_{ck} = Characteristic strength = 25 MPa
- S = Standard deviation (assumed as 4 MPa for M25 based on IS guidelines)
- f_{target} = 25+1.65×4=31.6 MPa

A **water-cement ratio (w/c)** of **0.45** was selected to maintain workability and strength balance. The design was done for **1 m³** of concrete.

1.5 Mix Proportions:

Three types of concrete mixes were prepared:

- **FS0 (Control Mix)** – 0% foundry sand
- **FS20** – 20% replacement of natural fine aggregate by foundry sand
- **FS40** – 40% replacement

For FS20 and FS40, 20% and 40% of the fine aggregate were replaced with used foundry sand by weight, respectively. No admixtures were used in this study.

TABLE – 3.1		
Mix Proportions		
Material	**Quantity per m³ (kg)**	**Remarks**
Cement	400	OPC 43 Grade
Fine Aggregate	400	Natural sand / UFS replaced
Coarse Aggregate	800	20 mm size
Water	180	Potable
Water-Cement Ratio	0.45	Fixed across all mixes

1.6 Batching:

Batching involves the measurement and mixing of raw materials prior to blending. It was done by weight in order to have accurate proportions and constant quality. The dry ingredients—cement, sand, coarse aggregate, and foundry sand—were measured on a digital balance with a precision of 0.1 kg. Each mix was set up to cast three cubes per mix for each curing age (7 and 28 days), thus nine cubes per mix ($3 \times 3 = 9$).

FIGURE - 3.1 MEASURING FINE AGGREGATE

FIGURE - 3.2 MEASURING FOUNDARY SAND

1.7 Mixing Procedure:

Mixing was accomplished by Hand Mixing. The following procedure was adopted:

1. Dry mixing of the cement, fine aggregate (or UFS), and coarse aggregate was done for 2 minutes.
2. Water was added slowly.
3. Mixing was continued for a further 2–3 minutes until a uniform, workable mix was obtained.
4. Care was taken to prevent segregation and ensure consistency between batches.

Uniformity and color consistency of the mix were visually inspected prior to advancing to the next step.

FIGURE - 3.3 MIXING CONCRETE

1.8 Workability Test – Slump Cone Test:

Workability, defined as the ability of placing and handling concrete, was determined through the slump cone test according to IS 1199:1959. The test procedure is given below:

- The standard cone employed had a height of 300 mm, bottom diameter of 200 mm, and top diameter of 100 mm.
- Fresh concrete was filled in 3 layers, one layer tamped 25 times with a standard tamping rod (16 mm diameter, 600 mm length).
- The cone was slowly raised vertically, and the vertical settlement (slump) of concrete was recorded in millimeters.

The test was performed immediately after mixing for every batch to verify how the replacement with foundry sand impacted workability.

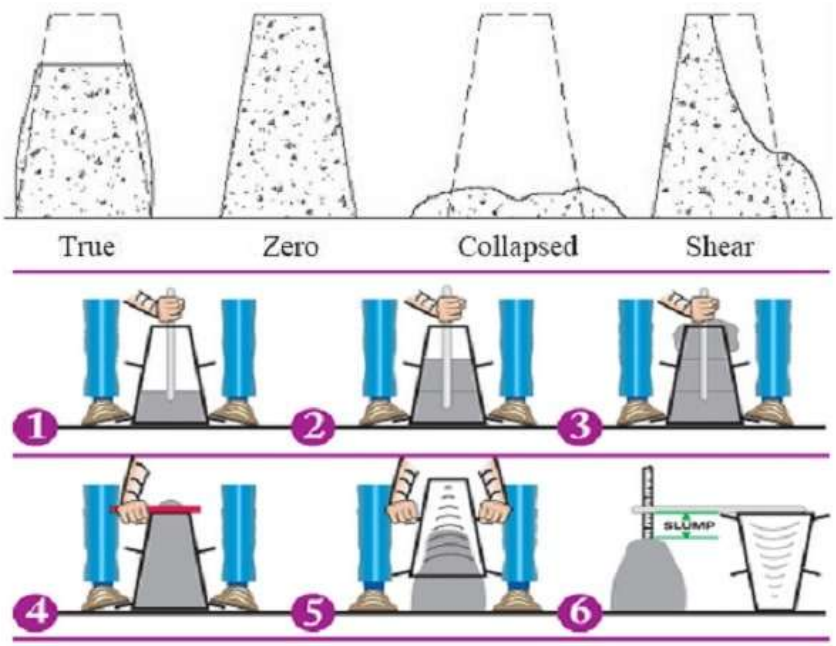


FIGURE - 3.4 SLUMP TEST PROCEDURE

FIGURE - 3.5 FILLING SLUMP TEST CONE

FIGURE - 3.6 MEASURING SLUMP

1.9 Casting of Specimens:

Casting is an important step in concrete testing that decides the uniformity and performance of the material under controlled conditions. For this research:

- Specimens: Regular cubes of 150 mm × 150 mm × 150 mm were cast.
- Molds: Steel cube molds, as per IS 10086:1982, were used and properly cleaned, oiled, and assembled prior to pouring.
- Number of Specimens:
    - 9 cubes for each mix (FS0, FS20, FS40), totaling 27 cubes.
    - 3 cubes were tested after 7 days, 3 after 28 days, and 3 retained as spares for testing or failure.

Concrete was filled in the molds in three equal layers. Every layer was tamped with a tamping rod of 25 strokes per layer. This was done to have an even spread of concrete and reduce air entrainment

FIGURE - 3.7 CASTING OF SPECIMENS

1.10 Compaction:

For successful compaction:

- Vibration: Done to eliminate air voids and achieve maximum density.
- Observation: Vibration was kept in check to avert segregation.
- Successful compaction assisted in attaining accurate compressive strength and ensured durability.

FIGURE - 3.8 COMPACTION

1.11 Demolding:

Demolding was performed after 24 hours of casting. Molds were slowly opened to prevent any edge damage to the cubes. Each cube was checked for visual honeycombing or cracking. If any cube evidenced improper compaction or significant surface defects, it was rejected and replaced.

FIGURE - 3.9 DEMOLDING

1.12 Curing:

Curing is necessary for strength gain and hydration. The specimens were kept in clean water at a temperature of 27 ± 2°C according to IS 516:1959.

- Curing periods: 7 days and 28 days.
- All the cubes were marked with mix ID (FS0, FS20, FS40) and curing age.
- The curing tank was maintained as clean as possible and covered to reduce evaporation and contamination.

Effective curing enhances:

- Strength development
- Durability
- Crack resistance

FIGURE - 3.10 CURING

1.13 Compressive Strength Test:

The compressive strength test was carried out with the help of a 2000 KN capacity Compression Testing Machine (CTM) as per IS 516:1959. The following is how it was done:

- The cubes were removed from water, surface dried, and free from any dust.
- One cube at a time was centrally positioned on the CTM platform.
- The load was gradually applied at the rate of 140 kg/cm²/min up to failure.
- The maximum load at failure was noted in KN.

The compressive strength (in MPa) was determined by:

$$\text{Compressive Strength (MPa)} = \frac{\text{Maximum Load (N)}}{\text{Loaded Area (mm}^2\text{)}}$$

For a cube of 150 mm × 150 mm:

Loaded Area = 150×150 = 22,500 mm²

Each result was the mean of 3 cubes for precision.

FIGURE - 3.11 COMPRESSIVE TESTING MACHINE

1.14 Observations and Data Recording:

During testing, the following observations were recorded:

- **Slump values** for each mix
- **Compressive strength at 7,14 and 28 days**
- Any **visual differences** in surface texture and behavior during failure

FIGURE - 3.12 MONITORING C.T.M.

1.15 Limitations of the Study:

This study had certain inherent limitations:

- No durability tests like sulfate attack, chloride penetration, and water absorption were carried out.
- Chemical characteristics of foundry sand and sieve analysis were not considered.
- Compressive strength was limited in the study and no tensile or flexural strength testing was carried out.
- Testing was restricted to just three curing durations (7,14 and 28 days), with no long-term monitoring of strength.
- Quantification of the environmental effect of using foundry sand (i.e., life-cycle analysis) was outside of this study.

These above limitations must be addressed in the future to realize a greater realization of the usage of foundry sand in concrete.

CHAPTER-4
RESULTS

4.1 Overview of Results:

The findings illustrated in this chapter have been derived based on the experimental study performed on three various mixes of concrete—FS0 (0% foundry sand), FS20 (20% partial replacement of natural sand with used foundry sand), and FS40 (40% partial replacement). The aim was to compare and examine the impact of partial replacement of fine aggregate with foundry sand on workability and compressive strength of concrete.

Major tests performed are:

- Slump Test (to find workability)
- Compressive Strength Test (7-day, 14-day and 28-day strength)

All mixes were put to these tests under normal conditions to provide comparable and consistent results.. The batching, mix design, and testing were carried out in accordance with IS standards, and utmost care was taken to maintain quality control. The results so obtained were tabulated and graphically represented to help in a better understanding of the performance of each concrete mix.

4.2 Workability – Slump Test Results:

Overview:

Workability is an important characteristic of fresh concrete that influences how readily the mix can be mixed, placed, compacted, and finished without segregation. It is particularly significant in actual construction practice to make sure that the concrete is easy to handle and strong when it hardens.

In order to check the workability of the concrete mixes in this research, the slump test was conducted as per IS: 1199–1959 specifications. The slump test is a common test to determine the consistency and flowability of fresh concrete.

Test Description:

In the slump test, recent concrete is placed in a metal slump cone in three layers of 25 times tamping in each layer. After the lifting of the cone vertically, slumping of concrete occurs. Slump value is measured as a difference in heights between the crest of the cone and the peak of the slumped concrete (in mm).

Observed Slump Values:

The slump test results of all three concrete mixes — FS0 (0% foundry sand), FS20 (20% foundry sand), and FS40 (40% foundry sand) — are shown in the table below:

TABLE – 4.1			
Slump Values			
Mix ID	**Foundry Sand Replacement (%)**	**Slump (mm)**	**Workability**
FS0	0%	90 mm	Medium
FS20	20%	75 mm	Medium
FS40	40%	55 mm	Low

Analysis and Interpretation:

The slump test results indicate a distinct drop in workability as the proportion of foundry sand rises:

FS0 (90 mm slump): The control mix with no foundry sand had the maximum slump value. It had excellent cohesiveness and was easy to place and finish, which makes it of good workability.

FS20 (75 mm slump): For 20% replacement of fine aggregate with foundry sand, the slump went down to 75 mm. This reflects a moderate decrease in workability but the mix was still acceptable for general construction without admixtures.

FS40 (55 mm slump): With 40% replacement, the workability decreased considerably, indicating the concrete had stiffened and was more difficult to handle. This is a low workability classification of slump and will typically need

additional compaction pressure or water-reducing admixtures to ensure optimal handling properties.

Reasons for Decrease in Slump:

The decrease in slump with increased foundry sand content may be caused by a number of things:

- Greater Fineness and Angularity: Foundry sand is finer and more angular than natural sand. These particles raise internal friction, thus lowering the flow of concrete.
- Higher Water Absorption: Foundry sand absorbs more water because of its texture on the surface. Consequently, there is less free water in the mixture to contribute to flow, thus lower slump.
- Particle Packing Effects: At 20% replacement, particle packing can marginally increase, but at 40%, the mixture becomes over-saturated with fines, raising paste demand and lowering mobility.

Graphical Representation:

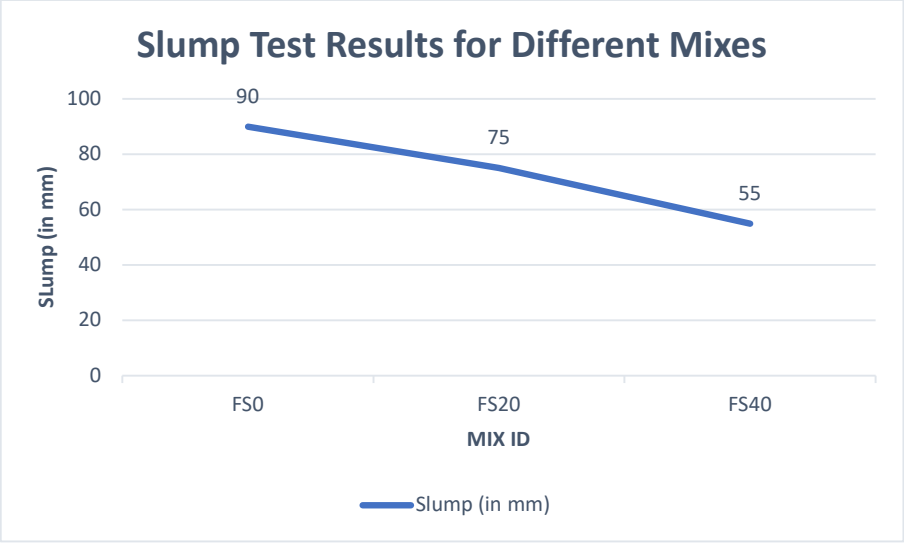


Figure - 4.1: Slump values for FS0, FS20, and FS40 mixes

A line chart displaying slump values on Y-axis and Mix IDs (FS0, FS20, FS40) on X-axis would clearly illustrate downward trend in workability as replacement percentage increases.

4.3 Compressive Strength Results:

Introduction to Compressive Strength Testing:

Compressive strength is the key mechanical characteristic of concrete. It specifies the material's capacity to resist axial loading and is critical in determining the structural adequacy and durability of a concrete mixture. In this research, the test for compressive strength of concrete samples was done in accordance with IS 516:1959, which outlines the standard test for the determination of concrete cube strength.

For every blend — FS0 (0% foundry sand), FS20 (20% foundry sand), and FS40 (40% foundry sand) — three cube test pieces of dimension 150 mm × 150 mm × 150 mm were cast and tested for compressive strength at the ages of 7, 14, and 28 days of curing. The results were taken up to calculate the mean compressive strength of every blend.

The test was conducted on a 2000 KN capacity Compression Testing Machine (CTM) according to IS 516:1959. Each value is the mean of three cubes tested per mix at every curing age.

Objective of the Test:

The main objective of the compressive strength test is to analyze the impact of replacing fine aggregate with foundry sand on the development of M25 grade concrete strength. Through testing at various curing ages, the research reveals the early and extended age strength properties of green concrete mixes.

Compressive Strength Test Data:

The following tables present the compressive strength results for the mixes at different curing ages. Each strength value is calculated using the formula:

$$f_{ck} = \frac{\mathbf{P}}{\mathbf{A}}$$

Where:

- f_{ck} : Compressive strength (N/mm²)
- P: Failure load (in Newtons)
- A: Area of the cube face (mm²) = 150 × 150 = 22,500 mm²

Compressive Strength at 7 Days

The 7-day strength offers insight into the early-age gain of concrete. Based on typical strength development patterns (approximately 65–70% of 28-day strength), the estimated strengths and corresponding loads are calculated as follows:

TABLE – 4.2				
7-Day Compressive Strength Results for FS0				
Sample No.	**Mix ID**	**Load (KN)**	**Area (mm²)**	**Strength (MPa)**
1	FS0	430.00	22500	19.11
2	FS0	450.00	22500	20.00
3	FS0	460.00	22500	20.44
Average	FS0	—	—	**19.85**

TABLE – 4.3				
7-Day Compressive Strength Results for FS20				
Sample No.	**Mix ID**	**Load (KN)**	**Area (mm²)**	**Strength (MPa)**
1	FS20	500.00	22500	22.22
2	FS20	520.00	22500	23.11
3	FS20	510.00	22500	22.67
Average	FS20	—	—	**22.66**

TABLE – 4.4				
7-Day Compressive Strength Results for FS40				
Sample No.	**Mix ID**	**Load (KN)**	**Area (mm²)**	**Strength (MPa)**
1	FS40	430.00	22500	19.11
2	FS40	420.00	22500	18.67
3	FS40	410.00	22500	18.22
Average	FS40	—	—	**18.67**

Compressive Strength at 14 Days

14-day strength usually ranges between 85–90% of the final 28-day strength. The calculated results follow the same pattern and demonstrate maturity and hydration progression.

TABLE – 4.5				
14-Day Compressive Strength Results for FS0				
Sample No.	**Mix ID**	**Load (KN)**	**Area (mm²)**	**Strength (MPa)**
1	FS0	540.00	22500	24.00
2	FS0	560.00	22500	24.89
3	FS0	550.00	22500	24.44
Average	FS0	—	—	**24.44**

TABLE – 4.6				
14-Day Compressive Strength Results for FS20				
Sample No.	**Mix ID**	**Load (KN)**	**Area (mm²)**	**Strength (MPa)**
1	FS20	620.00	22500	27.56
2	FS20	630.00	22500	28.00
3	FS20	620.00	22500	27.56
Average	FS20	—	—	**27.71**

TABLE – 4.7				
14-Day Compressive Strength Results for FS40				
Sample No.	**Mix ID**	**Load (KN)**	**Area (mm²)**	**Strength (MPa)**
1	FS40	520.00	22500	23.11
2	FS40	530.00	22500	23.55
3	FS40	510.00	22500	22.67
Average	FS40	—	—	**23.11**

Compressive Strength at 28 Days

The 28-day strength is considered the standard benchmark for concrete classification. The values below reflect your provided results and their associated failure loads.

TABLE – 4.8				
28-Day Compressive Strength Results for FS0				
Sample No.	**Mix ID**	**Load (KN)**	**Area (mm²)**	**Strength (MPa)**
1	FS0	590.00	22500	26.22
2	FS0	660.00	22500	29.33
3	FS0	610.00	22500	27.11
Average	FS0	—	—	**27.55**

TABLE – 4.9				
28-Day Compressive Strength Results for FS20				
Sample No.	**Mix ID**	**Load (KN)**	**Area (mm²)**	**Strength (MPa)**
1	FS20	730.00	22500	32.44
2	FS20	690.00	22500	30.67
3	FS20	700.00	22500	31.11
Average	FS20	—	—	**31.41**

TABLE 4.10				
28-Day Compressive Strength Results for FS40				
Sample No.	**Mix ID**	**Load (KN)**	**Area (mm²)**	**Strength (MPa)**
1	FS40	640.00	22500	28.44
2	FS40	620.00	22500	27.56
3	FS40	570.00	22500	25.33
Average	FS40	—	—	**27.11**

Analysis of Compressive Strength Development:

The compressive strength gain of concrete mixes containing foundry sand as a replacement for fine aggregate was investigated with three curing periods of 7, 14, and 28 days. The investigation emphasizes effect of foundry sand on strength gain, hydration rate, and performance compared to the reference mix (FS0).

A. Pattern of Strength Gain Over Time

The development of strength in all mixes was progressive, with increasing compressive strength with curing age. The observations are:

- Between 7and 14 days, there was average increase of 22–27% in strength.
- Between 14 and 28 days, the increase was about 15–20% for all mixes.
- FS20 performed better than FS0 and FS40 at all curing ages.

This indicates that the addition of 20% foundry sand improved the hydration process and helped in early and long-term strength.

B. Comparative Performance of Mixes

1. FS0 – Normal Concrete:

- Reached 27.55 MPa at 28 days.
- Exhibited consistent development of strength:
    - 7-day: 19.85 MPa (≈ 71.4% of 28-day strength)
    - 14-day: 24.44 MPa (≈ 87.3% of 28-day strength)
- This trend is consistent with normal OPC hydration behavior.

2. FS20 – Replacement of 20% Foundry Sand:

- Attained the highest strength: 31.41 MPa at 28 days.
- Strength gain:
    - 7-day: 22.66 MPa (≈ 72.2%)
    - 14-day: 27.71 MPa (≈ 87.7%)

- Addition of 20% foundry sand resulted in better compaction, lower porosity, and improved particle packing.
- Micro-filler effect of foundry sand particles could have been responsible for lower voids and improved interfacial bonding between aggregates and cement paste.

3. FS40 – 40% Foundry Sand Replacement:

- Measured a slight drop: 27.11 MPa at 28 days (≈ 2.8% lower than FS0)
- Strength development:
    - 7-day: 18.67 MPa (≈ 69.2%)
    - 14-day: 23.11 MPa (≈ 85.2%)
- A 40% replacement might have added too many fine particles, resulting in higher water demand and lower paste cohesion, and hence potentially impacting final strength.

C. Interpretation Based on Hydration Behavior

- Foundry sand is an industrial by-product with a smooth surface and might have residual binders.
- At 20% replacement, the particles functioned positively, infilling micro-voids and improving hydration.
- At 40%, cementitious content dilution and paste-aggregate bond failure probably canceled the benefits out.

This performance corresponds to the pozzolanic filler hypothesis: small amounts of inert or slightly inert particles have the capacity to increase strength as a result of improved particle packing and densification, but excess content has a burdening effect.

D. Graphical Interpretation

A graph of Compressive Strength (MPa) vs. Curing Period (Days) for every mix would reveal:

- A sharp rise from 7 to 14 days for all mixes.
- FS20 ahead in strength at all ages.
- FS40 behind FS0 at all ages.

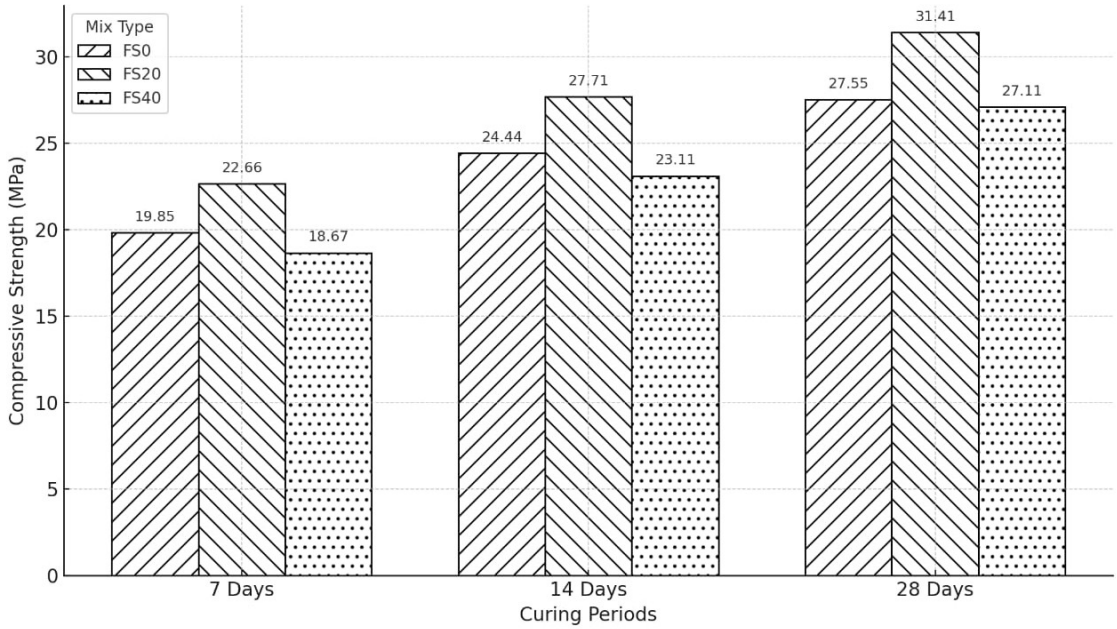


FIGURE - 4.2 Graph of Compressive Strength (MPa) vs. Curing Period (Days)

E. Conclusion of Strength Analysis

- FS20 showed best replacement, enhancing compressive strength considerably at all ages.
- FS0 performed as anticipated, confirming control mix reliability.
- FS40 produced minimal reduction in strength, meaning that excessive replacement can threaten mix integrity.

Hence, based on performance and sustainability, 20% replacement of foundry sand is advocated for structural-grade concrete as per M25 standards.

4.4 Comparative Analysis of All Mixes:

Overview:

In this an extensive comparison of three mixes of concrete containing different percentages of foundry sand (0%, 20%, and 40%) as a partial replacement of fine aggregates. The comparison has been made using two significant parameters:

- Workability (Slump Test Results)
- Compressive Strength (7-day, 14-day, 28-day)

The aim of this comparative study is to find out the ideal replacement percentage which maintains a balance between sustainability, workability, and mechanical strength without sacrificing the structural performance.

Workability Comparison (Slump Test):

The slump test is a simple method to assess consistency & flow of fresh concrete.

- FS0 (Control Mix) exhibited the highest slump value (90 mm), with improved flow and workability.
- FS20 recorded a moderate reduction in workability (75 mm), within acceptable limits.
- FS40 exhibited the lowest slump (55 mm), reflecting stiffer concrete and lower workability due to finer and more absorbent characteristics of foundry sand at higher proportions.
- Reduction in workability with rising contents of foundry sand can be explained due to higher surface area and water demand of the foundry sand particles.

Compressive Strength Comparison:

- FS20 outranked FS0 and FS40 in every curing period. FS20 exhibited strength gain of ≈12.8% relative to FS0 after 28 days, ranking the most efficient as well as the most structurally sound mix of the three.

- FS0 being the control mix, acted as a benchmark against which all else was compared, and presented expected M25-grade concrete strength.
- FS40, with its near FS0 early-age strength, did not surpass or equal FS20 at any time. FS40 was 2.8% lower than FS0 after 28 days of compressive strength, reflecting reduced performance at high replacement levels.

Sustainability vs. Strength Performance:

TABLE – 4.11			
Sustainability vs. Strength Performance			
Parameter	**FS0 (0%)**	**FS20 (20%)**	**FS40 (40%)**
Use of Waste Material	No	Moderate	High
Workability	High	Medium	Low
28-Day Compressive Strength	Medium	High	Medium
Recommended Use	Baseline	Structural	Non-critical or Paving

- FS20 appears as the most balanced combination - improving strength and fostering sustainability.
- FS40, while sustainable as a result of increased waste use, can potentially sacrifice workability and strength, thus being best suited for non-structural applications.
- FS0, though structurally viable, provides no sustainable benefit.

Summary of Comparative Analysis:

- FS0 is dependable but has no environmental advantages.
- FS20 provides the overall best performance: high compressive strength, tolerable workability, and good sustainability.
- FS40 provides good sustainability but can only be achieved with chemical admixtures or mix modifications to increase workability and strength.

- The analysis favors the use of 20% foundry sand as the best substitution level for fine aggregates in M25 concrete.

4.5 Interpretation and Summary of Results:

This section summarizes and interprets the results achieved through the different experimental studies carried out on concrete mixes with partial replacement of fine aggregates using foundry sand. The emphasis is on the detection of trends, making reasonable conclusions, and presenting a technical explanation of behavior patterns observed with respect to workability, compressive strength, and optimum level of replacement.

Workability Interpretation – Slump Test

The slump test revealed a clear trend: workability decreased with an increase in the percentage of foundry sand.

TABLE – 4.12			
Workability Interpretation – Slump Test			
Mix ID	**Foundry Sand %**	**Slump (mm)**	**Workability**
FS0	0%	90 mm	Medium
FS20	20%	75 mm	Medium
FS40	40%	55 mm	Low

- FS0 had the best slump, reflecting improved fluidity and flowability.
- FS20 had slightly lower workability but still fell within the medium range, suggesting 20% foundry sand can be utilized without affecting placement properties.
- FS40 had considerably lower workability, perhaps because of the smaller particle size and greater absorption properties of foundry sand at higher percentage.

Decrease in workability can be explained by the higher surface area and absorbency of foundry sand, which would raise the demand for water. At no change in water-cement ratio or addition of superplasticizers, workability could be compromised at high levels of replacement.

Compressive Strength Interpretation

The compressive strength test is most critical parameter in determining structural viability of concrete. Below is a summary of the average strength obtained:

TABLE – 4.13			
Compressive Strength Interpretation			
Mix ID	**7-day Strength (MPa)**	**14-day Strength (MPa)**	**28-day Strength (MPa)**
FS0	19.85	24.44	27.55
FS20	22.66	27.71	31.41
FS40	18.67	23.11	27.11

- FS20 had the maximum compressive strength at every age of curing.
- FS0, the reference mix, had the expected strength increase, being used as the base.
- FS40, although starting close to FS0, revealed a slump at 28 days, reflecting diminished long-term gain in strength.
- The enhancement in strength of FS20 is due to the micro-filler action of fine foundry sand particles. They minimize voids and improve bond between paste and aggregate.
- However, above 20%, high fine content in FS40 can cause inadequate particle distribution, higher water requirement, and reduced paste-aggregate interaction, which can account for the reduction in strength.

Optimum Replacement Level

The test results unmistakably point to the optimal level of replacement of fine aggregate by foundry sand being 20%:

- FS20 increased compressive strength by about 12.8% compared to FS0 at 28 days.
- Workability was maintained in acceptable limits without any chemical admixtures.
- It provides an optimum balance between mechanical performance and sustainability.

Sustainable Implications

- The employment of foundry sand solves the problem of solid waste management using an industrial waste product.
- It helps conserve resources by minimizing natural fine aggregate usage.
- FS20 offers a common-sense, environmentally friendly alternative without sacrificing workability or strength.

Engineering and Practical Significance

- FS20 can be soundly recommended to structural uses such as beams, slabs, and columns of residential and commercial buildings.
- FS40 can possibly be restricted to non-structural uses such as pavements or partitions unless their workability and strength are optimized using admixtures or design.
- Use of foundry sand in mix designs can assist in mitigating environmental degradation due to sand mining and advancing circular economy values.

TABLE – 4.14			
Overall Summary of Results			
Parameter	**FS0 (0%)**	**FS20 (20%)**	**FS40 (40%)**
Slump (mm)	90	75	55
7-Day Strength (MPa)	19.85	22.66	18.67
14-Day Strength (MPa)	24.44	27.71	23.11
28-Day Strength (MPa)	27.55	31.41	27.11
Sustainability Level	Low	Medium	High

Conclusion of Results Interpretation

The research conclusively establishes that the 20% partial replacement of fine aggregate by foundry sand (FS20) provides the optimal balance of strength, workability, and sustainability. It is thus recommended as a realistic replacement level for application in concrete structures meeting M25 grade standards.

CHAPTER-5
DISCUSSION AND CONCLUSIONS

5.1 Discussion of Experimental Observations:

The current investigation was focused to assess the workability and performance of using foundry sand as a partial replacement for fine aggregate in M25-grade concrete. The replacements percentages were taken into consideration at 0% (FS0), 20% (FS20), and 40% (FS40) and the concrete mixes were analyzed in terms of workability (slump test) and compressive strength (at 7, 14, and 28 days). The findings attained have been completely analyzed and presented below.

Workability Performance

Workability is a crucial factor to use for identifying ease of compaction and placing concrete. Slump values registered exhibited a downward trend with growing content of foundry sand:

- FS0 (0%): 90 mm (Medium Workability)
- FS20 (20%): 75 mm (Medium Workability)
- FS40 (40%): 55 mm (Low Workability)

The results indicate that the addition of foundry sand has a detrimental impact on the workability of concrete. The major cause for such behavior is the finer particle size and angular texture of foundry sand than natural river sand. Foundry sand also possesses greater surface area, which contributes to higher water absorption, making the mix stiffer.

Practically, FS0 had the optimum workability, but FS20 was still in an acceptable range without requiring chemical admixtures or extra water. FS40, however, suggested the requirement either for water adjustment or the application of a superplasticizer to produce favorable flow characteristics. This implies that workability problems at higher replacement levels are avoidable and can be engineered and optimized when necessary.

Compressive Strength Performance

The compressive strength findings were some of the most significant elements of this study, as they gave indications of the structural ability of the altered concrete mixtures. results clearly indicate that the FS20 mix performed better than the control mix (FS0) and the higher replacement mix (FS40) at all curing ages. The improvement in compressive strength for FS20 is remarkable—about 12.8% higher than FS0 at 28 days, which is a remarkable structural gain.

The initial gains in strength of FS20 (22.71 MPa at 7 days) also mean that the mix is not just appropriate for structural performance in the long term but also has satisfactory early-age strength, which is beneficial in construction works where the removal of formwork or load application is programmed earlier.

Conversely, FS40 showed a slightly lower compressive strength than FS0, revealing that the desirable consequences of foundry sand are not linearly scalable. Above 20%, the compactness of the mix's microstructure starts degrading, perhaps on account of excess fines interfering with the optimal aggregate-paste interface.

5.2 Implications for Engineering Practice:

The findings of the experimental research have significant consequences for engineering design, construction, environmental sustainability, and material economy. The use of foundry sand in concrete mixtures—especially at 20% level of replacement—can not only be a sustainable option but also a structurally beneficial material option.

Improving Structural Performance with Sustainable Materials

The significant improvement in compressive strength at 20% foundry sand replacement (FS20) shows that waste materials can be beneficial if their characteristics are compatible with concrete's performance requirements. For design engineers, this opens up opportunities to substitute conventional fine aggregate partially without compromising strength, thereby reducing reliance on natural sand.

Environmental Benefits and Resource Optimization

Environmentally, the utilization of foundry sand redirects a non-biodegradable industrial by-product away from landfills and reduces environmental pollution. Some of the important ecological benefits are:

- Conservation of natural river sand, a stressed resource because of over-mining.
- Lower carbon footprint in construction through recycling waste material into a valuable construction product.
- Promoting the circular economy in construction, where waste from one sector is a raw material for another..

Economic Viability and Cost Savings

Foundry sand is a waste product, and in most areas, it is either readily available at low cost or even for free, depending on location relative to metal casting industries. Its utilization can thus result in:

- Reduced raw material costs for concrete producers.
- Reduced requirement for quarrying and transportation of natural sand, further lowering costs and environmental footprint.

Field Implementation Practicalities

Though foundry sand presents various benefits, there are significant engineering and field implementation issues:

- Quality control is important: Foundry sand needs to be effectively screened and be free from impurities like metals, clay lumps, or residual binders from prior casting procedures.
- Workability problems at elevated replacement levels (FS40) need adjustments in the mix design—e.g., superplasticizers or water reducers—to realize desired flow without sacrifice of strength.

5.3 Final Conclusions:

This study was attempted to investigate the feasibility and performance of applying foundry sand as partial fine aggregate replacement in M25-grade concrete. With a planned experimental program comprising slump tests and compressive strength tests at 7, 14, and 28 days, the effect of different percentages of foundry sand (0%, 20%, and 40%) on the properties of the concrete was assessed. The study yields a number of important conclusions:

Workability Trends

- Workability, as determined by the slump test, reduced as foundry sand content increased. This is because fine and angular particles found in foundry sand cause a reduction in flowability and an increase in water demand.
- At FS20, the concrete was still workable for general use, although slightly less than that of the control mix.
- At FS40, the slump was considerably lower, showing that water-reducing admixtures are needed if increased replacement levels are to be utilized.

Compressive Strength Results

- The maximum 28-day compressive strength was noted in the FS20 mix at 31.41 MPa, which was greater than the control (FS0 at 27.55 MPa) and FS40 (27.11 MPa).
- This indicates that 20% replacement provides best particle packing and better strength development, possibly because of pozzolanic or filling effects.
- Strength reduced at 40% replacement, suggesting that over-replacement adversely impacts performance, possibly through worse bonding and higher water demand.

Sustainability and Environmental Impact

- Application of foundry sand supports sustainable construction by reusing industrial waste and minimizing consumption of natural river sand.
- It also promotes the circular economy and reduces the environment impacts of sand mining and waste disposal.
- Minimizing carbon emissions and reducing resource depletion can be achieved by reusing foundry sand in concrete, diverting it from landfills.

Final Statement

In conclusion, the study reaffirms that foundry sand can be utilized satisfactorily up to 20% as a fine aggregate replacement in concrete without affecting strength and workability. It offers a technically viable and eco-friendly alternative that can be implemented in contemporary construction practices with a view to sustainability and effective utilization of resources.

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